Economics of Health and Health Care
Lectures 6 and 7: Health insurance

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Outline – information problems

- Health insurance: why and how?
- Strategic use of information – adverse selection and market equilibrium
- The increasing monopoly effect & empirical evidence
- Small area variations
- Agency relationship
- Strategic use of information – supply-induced demand

Imperfect information

- Up to this moment, we have kept the assumption of perfect information
- Relaxing this assumption leads to:
  - Uncertainty - the role of **insurance**, for example, but also what patients know
  - Asymmetric information - and how it changes the market equilibrium - **moral hazard** and **adverse selection**
  - **Agency relationship** - delegation of decision and its implications

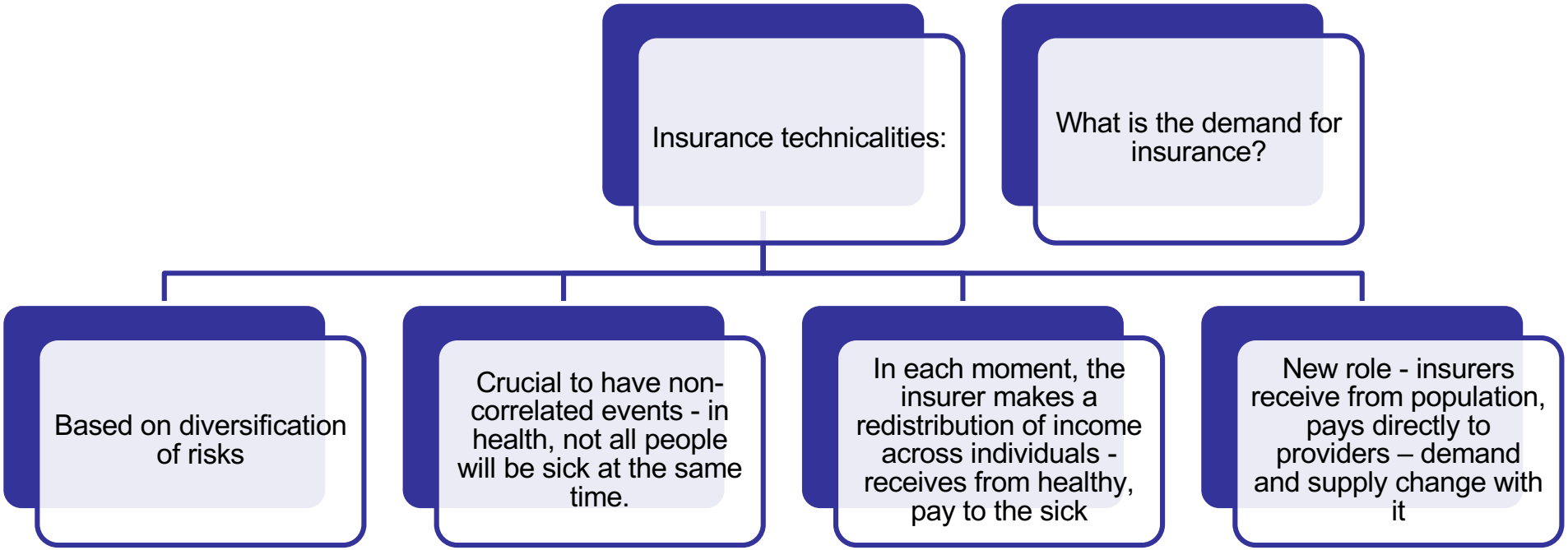
Health insurance

Main feature of health markets: uncertainty about the moment and value of medical expenditures that are needed

This uncertainty leads to demand of protection mechanisms - insurance

Insurance: transfer of financial responsabilities associated with future health expenditures to a third-party against a fixed payment.

Health insurance



Introduction

Two broad ways of organizing health insurance:

Private insurance - a company chargers insurance premia, usually according to risk characteristics of consumers

Public insurance - a public insurer (central fund, Government, etc…) - collects contributions (taxes or some other compulsory rule); the amount of contributions is typically determined by income level and not be risk characteristics.

While having private insurance is usually a voluntary decision, participation in public health insurance schemes is compulsory (exerted by the coercive action of the Government)

Several countries have also mandated private health insurance + regulation of insurance market

Can we explain these options?

The financing characteristics (and equilibrium) of a health system will depend on choices about public and private insurance

For the moment, let's focus on the common features to private and public health insurance

General principles that describe how the market for health insurance works

Demand for insurance

A consumer has insurance to reduce the variance of his payments under the set of possible contingencies:

The economic value of reducing that variance provides the margin for the insurance busines.

Another gain - access to health care without facing financial constraints

having the same income/consumption when sick and when not sick is better than having asymmetric consumption due to payment of health care expenditures

Key claims

In the absence of asymmetric information problems, full insurance is the best allocation

If there is moral hazard – people deciding demand for health care after insurance contract is set, people deciding on individual prevention measures – it is optimal to have less than full insurance. How much less depends on risk preferences and strength of moral hazard

If there is adverse selection – the private market will not provide enough insurance coverage.

Decision under uncertainty

Supporting these claims requires technical analysis - sketch of formal proofs

Consumers act as to maximize expected utility

Utility depends on the state of nature (sick / not sick)

Decision under uncertainty

- Utility function = v, with $v' > 0$, $v'' < 0$ - positive but decreasing marginal utility of consumption
- Consumers with equal income Y
- Patient has probability π of sickness with cost X

Expected utility

- $EU = \pi\, v(Y\text{-}X) + (1\text{-}\pi)\, v(Y)$ wihtout insurance
- With insurance pays premium P whether is healthy or sick, receives indemnity I if sick.
- Expected utility is: $EU = \pi\, v(Y\text{-}X\text{-}P\text{+}I) + (1\text{-}\pi)\, v(Y\text{-}P)$

Actuarially fair insurance

- Actuarially fair insurance = competitive market pricing
- zero expected value of financial flows with insurance company (competition)
- Premiums = claims paid: $P = \pi I$

Optimal choice

- The consumer selects the level of coverage, which is given by I
- If the consumer maximizes expected utility, the first-order condition is given by:

$$v'(Y-X+(1-\pi)I) = v'(Y-\pi I)$$

- marginal utility is the same in all states of Nature

Optimal choice

This condition only holds for

$$I=X$$

- The optimal choice is to have full insurance
- No variability in utility: $v(Y-\pi I)$

RESULT

In a competitive insurance market, with no distortions, the consumer has full insurance. It has for sure the expected value of his consumption.

In reality, we seldom see full insurance. Why?

- Real resources costs
- Market power
- Information problems - moral hazard and adverse selection.

Real resources costs

(not covered in class, leave it here for information)

- Price per unit of coverage: $P = p\ I,\ p > \pi$
- Solving the problem of maximization of expected utility, we now have

$$\frac{v'(Y - X - pI + I)}{v'(Y - pI)} = \frac{1-\pi}{\pi}\frac{p}{1-p}$$

- As $p > \pi$, it results

$$v'(Y-X-pI+I) > v'(Y-pI)$$

- Given the assumption of risk aversion, $v'' < 0$, it must be the case that:

$$I < X$$

- There is less than full insurance.

- This result is true for both private and public insurance.
- It depends on the real resources costs being a proportion of the coverage selected and not a fixed costs. If they are fixed costs, we recover the full insurance result.

Moral hazard

There is moral hazard when two parties (insurer and consumer) enter in a contract under symmetric information, but afterwards one of the sides (say, the consumer) does an action that cannot written in the contract and which changes the value of insurance.

Distinction:

ex-ante moral hazard	ex-post moral hazard.

Moral hazard

ex-ante moral hazard: when the action of the economic agent is done before realization of uncertainty

Ex-post moral hazard: when the action of the economic agent is done after realization of the state of nature (which has become known).

Assumption: actuarially fair prices.

What are the main results?

- With choice of quantity of care and no observability problems (can observe and verify everything relevant: insurance contract will have full insurance and specifies level of health care demanded
- With observability problems – **ex-post moral hazard – less than full insurance to control excessive demand of health care**

What are the main results?

- With prevention decisions – no observability problems – contract will have full insurance and specify how much prevention is done
- With observability problems – ex-ante moral hazard – less than full insurance to give incentives for prevention

Proving the claims: basic model

- Two states of nature: sick and healthy
- Being sick has disutility *B*, which can be reduced by consuming medical goods *x* in an amount *b(x)*
- The consumer may do prevention in amount *q* with cost *f(q)*:

$$U(Y-x) - \theta (B - b(x)) - f(q)$$

θ - Indicator of sickness (=1) or no (=0) (state-dependent utility)

Proving the claims: basic model

- The choice of x is ex-post moral hazard - the level of medical goods consumption is selected after the consumer learns he is sick.
- The choice of q is ex-ante moral hazard - the level of prevention care is selected before knowing whether the consumer is sick or not. The choice of this prevention effort may be difficult to include in the contract.

- Choice of x, $p(q)$ fixed, we can ignore B and $f(q)$ without loss of generality
- Solve by backward induction - first, characterize what happens when sick; then, characterize choice of insurance coverage, anticipating the choice of x in case of sickness

Ex-post moral hazard

- Being sick, without insurance, means a choice of maximizing utility subject to a behavior constraint (optimal choice of care demanded):

$$\max_x U(Y - x) + B(x)$$

$$U'(Y - x^*) = B'(x^*)$$

- Price of insurance:

$$P = p\ c\ x$$

- p - price per unit of coverage (equal to the probability of being sick for fair insurance)
- *1-c* - co-insurance rate: *c* is the share of cost paid by the insurance company (or central fund)

$$\max_x U(Y - P - (1-c)x) + B(x)$$

First-order condition:
note that *P* is fixed, it has been paid already

$$(1-c)U'(Y - P - (1-c)x) = B'(x)$$

Insurance premium given by

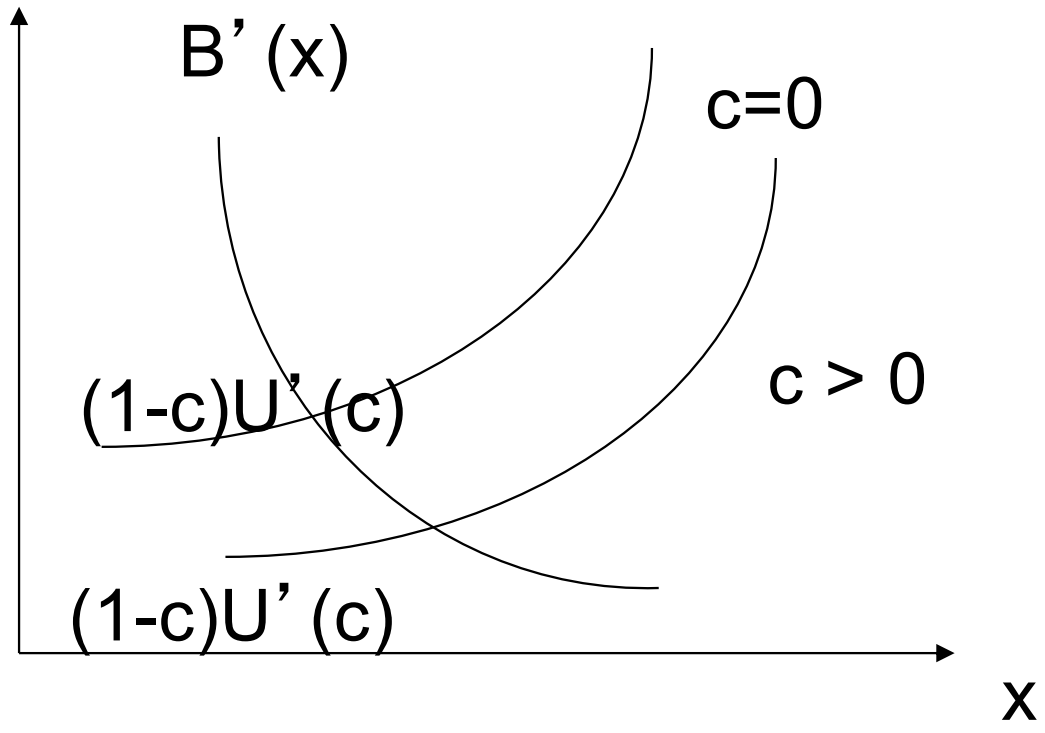
$$P = pcx$$

Equilibrium value characterized by

$$(1-c)U'(Y - xpc - (1-c)x) = B'(x)$$

By total differentiation

$$\partial x / \partial c > 0$$


B’(x)
c=0
c > 0
(1-c)U’(c)
(1-c)U’(c)
x

Optimal insurance contract - (x,c) are both contractible

$$\max_{\{x,c\}} V = pU(Y - pcx - (1-c)x) + pB(x) + (1-p)U(Y - pcx)$$

$$\frac{\partial V}{\partial c} = pU'(D)(-px + x) + (1-p)U'(S)(-px) = 0$$

$$\frac{\partial V}{\partial x} = pU'(D)(-pc - (1-c)) + pB'(x) - pc(1-p)U'(S) = 0$$

The first condition implies U' (D)=U' (S) which means full insurance. The second condition gives:

$$U'(y - px) = B'(x)$$

With moral hazard

- The choice of *x* is done after the contract is signed and *c* is known.
- Take this choice to be given by *x=x(c)*
- The consumer's problems is

$$\max_{c} V$$

$$s.t.$$

$$P = cpx(c)$$

$$x = x(c)$$

- From this problem, we have a first-order condition for the optimal choice of *x*

$$(1-c)U'(Y-px+(1-c)(1-p)x)=B'(x)$$

- The comparison with the optimal level is not clear.
- For c=1, equal coverage, it is easy to see medical care consumption exceeds the value in the socially optimal contract.

- On the other hand, for $c=0$, no insurance, the level of medical care is smaller than the socially optimal level.
- None of the two limiting cases leads to the socially optimal choices

- For each *c* exists a price effect in the demand for medical care - it stimulates demand.
- But there is an income effect through the insurance premium - which decreases the demand for medical goods
- For *c* close to 0, the income effect dominates; for *c* close to 1, the price effect dominates.

Optimal insurance under moral hazard

- If there is full insurance, the price at the moment where consumption is done is zero (to the consumer) - there will be an excessive consumption (this effect was already known)
- It implies that the second-best contract must have a positive co-insurance rate.

Insurance under ex-post moral hazard

- The optimal co-insurance rate results from the balance of risk protection and avoid excessive consumption.
- This is an economic justification for the existence of co-payments and “user charges” (fee paid to access health care).
- It holds under private or public insurance.

Ex-ante moral hazard

- Prevention action - taken before knowing the state of nature but influencing it (eg. driking before driving)
- $p=p(q)$, x fixed
- The probability of sickness is decreasing the in the amount of prevention effort.

Ex-ante moral hazard without insurance

$$\max_q V = p(q)U(Y - x) + (1 - p(q))U(Y) - f(q)$$

The first-order condition is given by

$$\frac{\partial V}{\partial q} = p'(q)[U(Y - x) - U(Y)] - f'(q) = 0$$

We have the condition of marginal benefit equal to marginal cost.

Optimal insurance with prevention

$$\max_{\{q,c\}} V = p(q)U(Y - x(1-c) - cp(q)x) + (1 - p(q))U(Y - p(q)xc) - f(q)$$

The first - order conditions are

$$\frac{\partial V}{\partial q} = p'(q)[U(D) - U(S)] + p(q)U'(D)(-cxp') + (1-p)U'(S)(-cxp') - f' = 0$$

$$\frac{\partial V}{\partial c} = px(1-p)U'(D) + (1-p)(-px)U'(S) = 0$$

- From this last condition, we see again that absent information issues the optimal insurance contract involves full insurance.

Second-best contract

For the consumer

$$\max_q V = p(q)U(Y - P - (1-c)x) + (1 - p(q))U(Y - P) - f(q)$$

$$\frac{\partial V}{\partial q} = p'(q)(U(D) - U(S)) - f'(q) = 0$$

which defines $q(c)$ with q decreasing in c

For the insurance company

$$\max_{\{q,c\}} V = p(q(c))U(D) + (1 - p(q(c))U(S) - f(q(c))$$

the first - order condition can be written as:

$$(1-p)px[U'(D) - U'(S)] = q' xcp'[pU'(D) + (1-p)U'(S)]$$

From this *U'(D) > U'(S)*, which is only possible for c < 1

Insurance under ex-ante moral hazard

- The existence of ex-ante moral hazard also justifies insurance contracts with less than full insurance.
- The consumer needs to bear some risk to be willing to incur in prevention costs.
- There is a balance to be made between risk protection and incentives for prevention.

differences in control of moral hazard

- public insurance - zero price at the point of consumption
- gives origin to extra consumption
- what we termed moral hazard effect
- private insurance - introducing a copayment, controls moral hazard

- for the same risk characteristics, less than full coverage in insurance allows for:
  - premium charged smaller than expected health expenditures
  - net gain, if premium charged by private parties is actuarially fair

Under moral hazard

Adverse selection

- The distortions in the insurance market so far have been the result of (non-contractible) actions taken after signing the contract.
- Information asymmetries before signing the contract may also have a large impact.
- We have seen the Akerlof argument related to adverse selection.
- Let's explore more completely the adverse selection issue.

Adverse selection

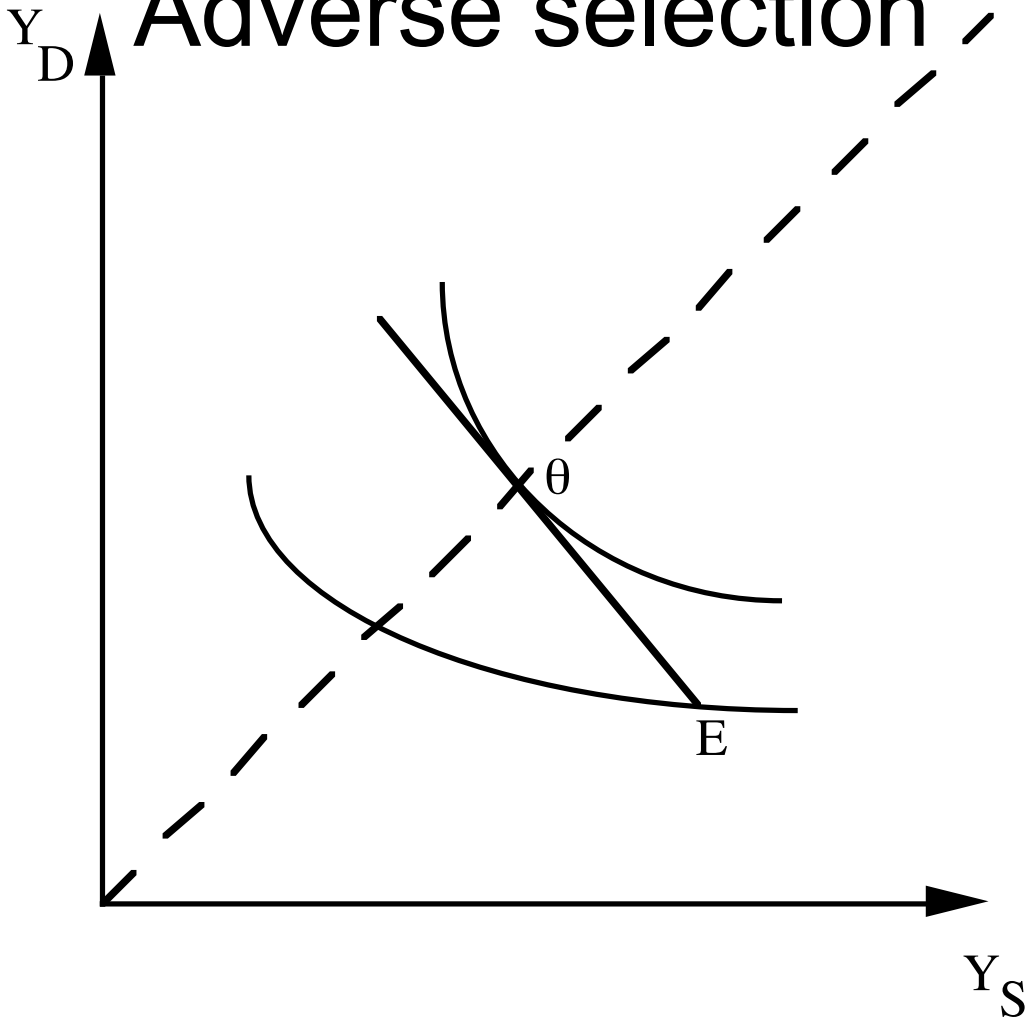
- The main difference is the existence of ex-ante asymmetric information,
- Say, two types of consumers/patients.
- each type of patient has a different probability of being sick.
- Such probability is only known by the consumer.
- What is the change in the insurance market?
- What is the equilibrium?

Adverse selection

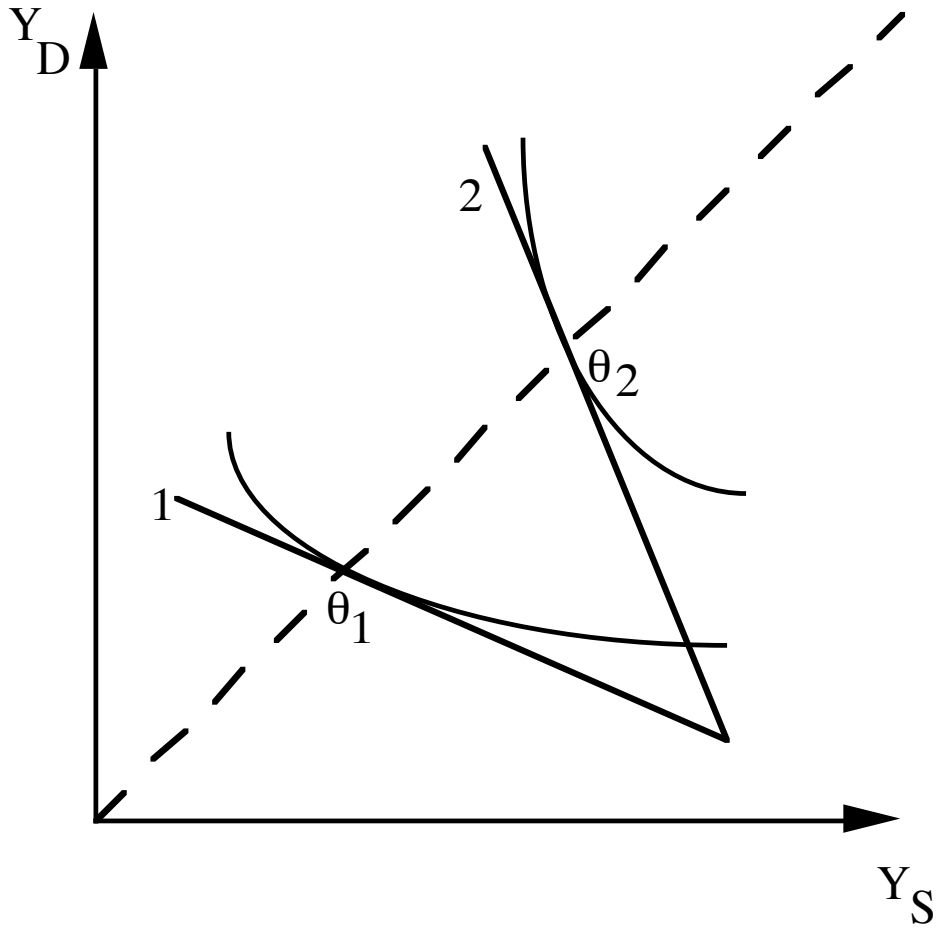
- Without adverse selection (and moral hazard), full insurance is optimal.
- Under actuarially fair insurance, consumers with higher risk/probability of being sick pay a higher insurance premium
- Graphical analysis uses space of consumption in each state of nature
- The insurance contract can be described as a payment made in the good state of nature (healthy) to receive a transfer in the bad state of nature (sickness).

Adverse selection

- Graphically

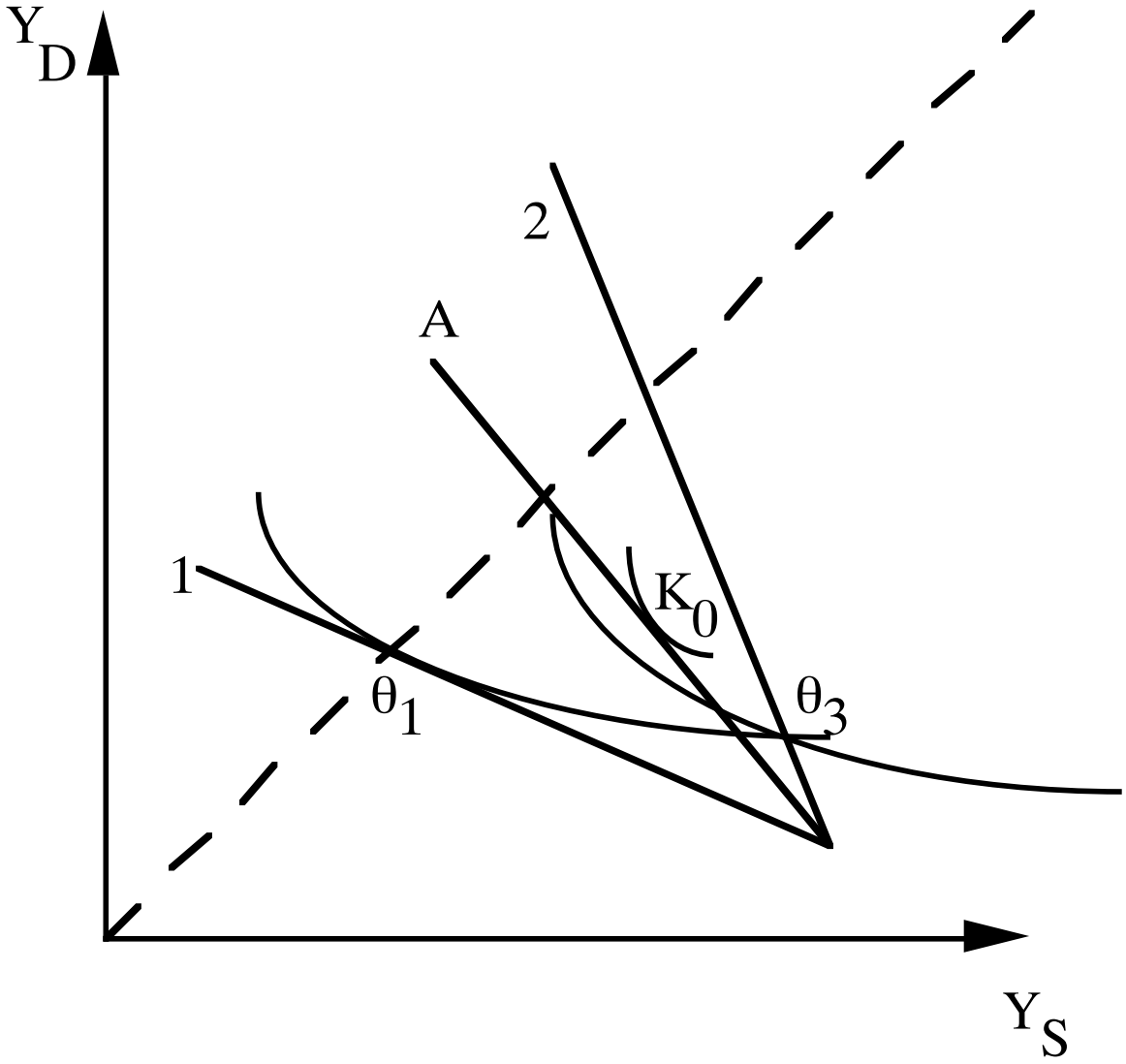


Full insurance for both types



Insurance contracts under adverse selection

- Assume now the insurance company is not able to distinguish whether a consumer has a high or low probability of sickness.
- Can it offer the same two insurance contracts and get it “right”?
- the answer is NO!
- Is it enough to have a single contract?


Y_D
2
A
1
K_0
θ_1
θ_3
Y_S

Optimal insurance and adverse selection

- A first important result, and a general one, is that a contract common to both types of consumers, in a competitive market, is not an equilibrium.
- It is always possible to offer another contract that attracts only the good consumers (low risk/probability of sickness) and gives positive profits.
- It is possible, under other assumptions, to recover the existence of a single contract: fixed costs, and/or expectations about reactions of other agens in the market.

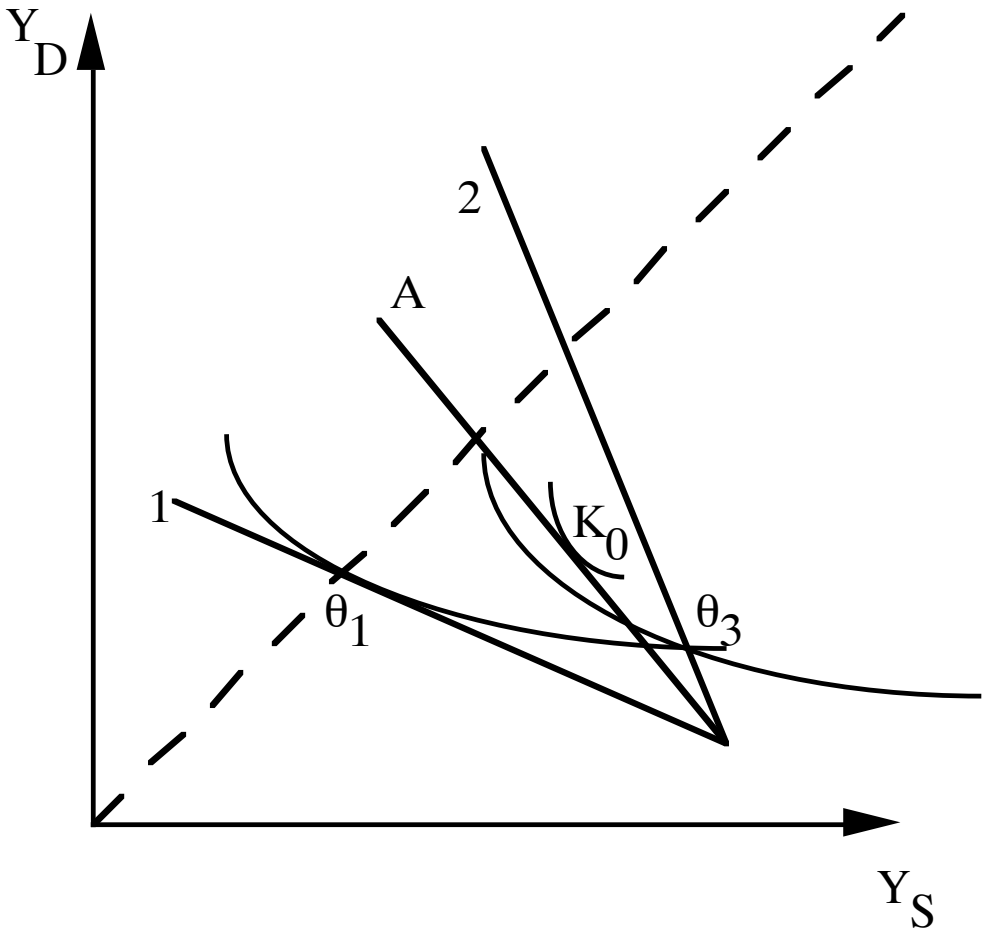
Optimal insurance and adverse selection

- Market equilibrium involves two contracts, one offered for each type of consumer.
- General characterization:
  - Consumers with the high probability of sickness obtain full insurance
  - consumers with low probability of sicjness have a contract distorting, receive less than full insurance.
  - the distortion arises from the need to avoid that “bad” risks select the other contract.

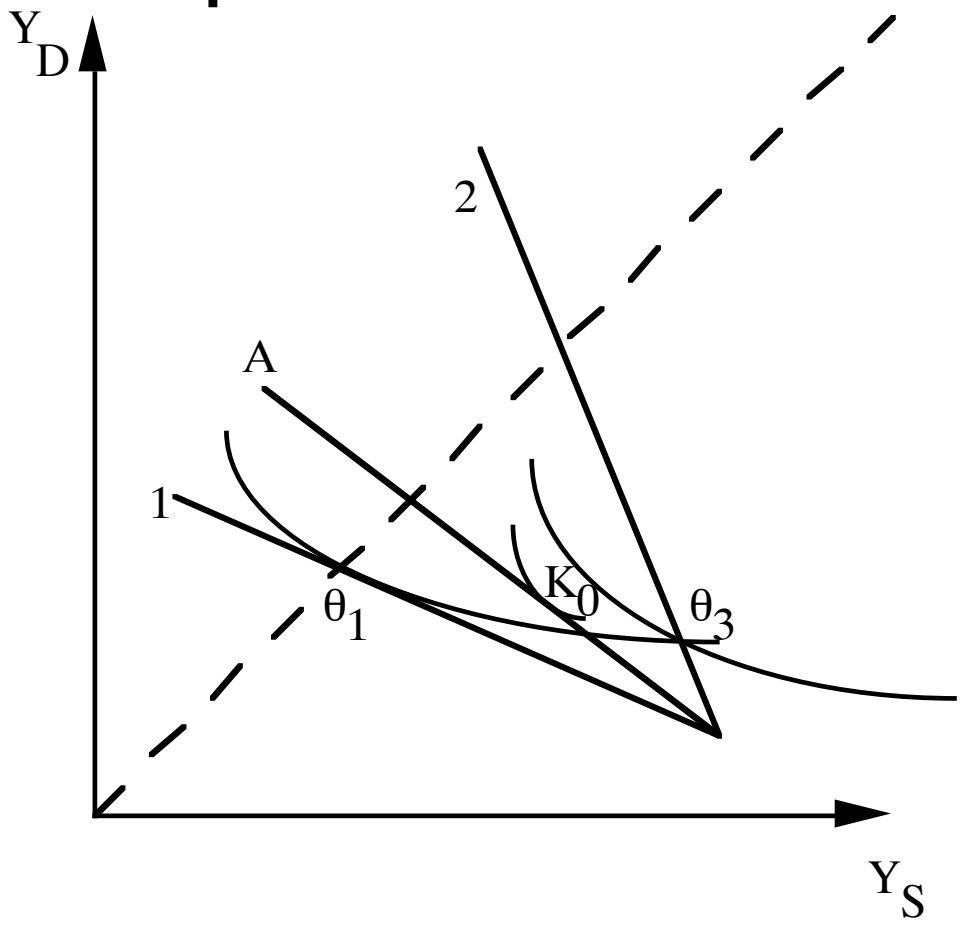
Optimal insurance under adverse selection

- Moreover, this equilibrium may not exist
- it depends on the proportion of “good” and “bad” risks in the population.
- An equilibrium may not exist, as the two contracts proposed may be “destroyed” by a single contract, if there is a sufficiently high number of good risks.

No equilibrium



Equilibrium exists



Insurance market equilibrium

- These results means that a private insurance market can be unstable.
- If it is stable, there are distortions that hurt the good risks.
- We cannot conclude that a public insurer is better, but this strongly suggests that the option between private and public supply is not has obvious as in other countries.
- the public insurer has the power to impose a pooling contract. But it is a monopoly, with other sort of costs.

Insurance market equilibrium

- The choice between private and public insurer also hinges on other elements:
  - what is the real significance of adverse selection in insurance markets?
  - what distortions arise in the alternative model of public insurer?
  - what properties have the two systems on equity grounds?
- European answer: favours the public insurance mechanism (or similar) in the financing of the health system.